

# Partition Draws, Not Aggregators: Decomposing Result Variance in Federated LoRA Fine-Tuning Under Non-IID Splits

JERRY ADAMS FRANKLIN<sup>1</sup>

<sup>1</sup>Independent Researcher (ORCID: <https://orcid.org/0009-0006-8470-8349>)

Corresponding author: Jerry Adams Franklin (e-mail: [jerry.adamsf@gmail.com](mailto:jerry.adamsf@gmail.com)).

**ABSTRACT** Federated low-rank adaptation (LoRA) comparisons often rest on a single non-IID partition draw, even when reported method gaps sit near run-to-run noise. We measure how much held-out loss variation the partition draw causes, how often few-draw rankings flip, and how many draws a given effect requires. In a pre-registered design of 204 runs at two scales (TinyLlama-1.1B-Chat and LLaMA-3.2-3B), with partition seeds separated from training seeds, we decompose the variance of FedIT, FFA-LoRA, and FLoRA on Dolly with Dirichlet label skew at  $\alpha=0.1$  and  $\alpha=0.5$ . On TinyLlama, partition and interaction shares are 0.508 and 0.469 against a 0.023 residual. On LLaMA-3.2-3B with ten draws the interaction dominates (0.992) and the partition share truncates to zero. Flips concentrate in the near-tied FedIT–FLoRA pair (three-draw paired flip probability 0.377 and 0.493), while gaps of 0.023 to 0.031 never reverse. Detecting that near-tie at 80% power needs 206 paired draws on TinyLlama and more than 1000 on LLaMA-3.2-3B; separable pairs need 3. Extending the 3B grid from six to ten draws reverses the top-two ordering and grows the interaction component 2.66-fold, so stopping at six would report the opposite ranking. At  $\alpha=0.5$  the composition is statistically unchanged while total variance shrinks by about an order of magnitude. We propose the Partition-Draw Reporting Protocol: report the number of partition draws and their seeds separately from training seeds, pair method comparisons on the same draws, and report the minimum detectable difference. We make no method recommendation.

**INDEX TERMS** federated learning, low-rank adaptation, non-IID partitioning, experimental variance, reproducibility, large language models

## I. INTRODUCTION

Federated LoRA methods for instruction tuning report method differences that are often small relative to the noise in a single evaluation. FFA-LoRA [1] publishes means over 20 runs with standard deviations; in the non-private setting the LoRA standard deviation on MNLI-matched (10.7) exceeds the mean gap to FFA-LoRA (3.02 points). FLoRA [2] compares to FedIT [3] with single reported MMLU [4] and MT-bench [5] values after 1 to 3 communication rounds, without reported seed replication, including a Llama/Alpaca MMLU margin of 0.44 points that the paper itself calls “marginal.” Those practices establish the motivating fact: published federated LoRA contrasts can sit inside, or only slightly above, ordinary run-to-run variation. What they do not establish is how much of that variation comes from the

non-IID partition draw itself.

In Dirichlet or category-based federated simulations, the “non-IID” condition is treated as a setting rather than a source of variance in the evaluations examined here: the Dirichlet construction itself [6] and the partition catalogues built on it [7] do so, and Section II gives benchmarks from 2025 and 2026 that do the same. In the evaluations cited above, training seeds are sometimes repeated while the partition seed is not. If held-out loss moves with the partition draw by as much as it moves with the training seed, then a ranking obtained from one or two partitions is not a property of the methods alone. The open question is quantitative: how large is the partition component, how often do few-draw rankings flip, and how many draws are needed for a target effect size?

We answer three pre-registered questions on FedIT, FFA-

LoRA, and FLoRA, using held-out next-token loss on Dolly as the response. All differences are in held-out next-token loss, where the unadapted base models score 2.121 and 2.169 and fine-tuned runs fall between 1.691 and 1.841. The design freezes the federation (ten clients, full participation, fifteen rounds, one local epoch), the LoRA configuration, and the analysis plan before any production run, and separates partition seeds from training seeds at two model scales.

- **RQ1.** At Dirichlet  $\alpha=0.1$ , what share of the variance comes from the partition draw, the partition-by-method interaction, and the training seed?
- **RQ2.** How often does an evaluation using  $k$  partition draws rank the three methods differently from the full-design ranking?
- **RQ3.** How many partition draws are needed to detect held-out loss differences of 0.005, 0.01, 0.02, and 0.05 at 80% power, for paired and unpaired designs?

An exploratory RQ4 on partition statistics is reported but not used for claims.

The design extension adds a second heterogeneity level and a larger partition count, which test whether the RQ1 composition and the RQ2 ranking behavior hold beyond the originally registered setting.

### A. ADVANCE OVER THE STATE OF THE ART

Prior federated LoRA evaluations either report multi-seed means without separating the partition draw from the training seed, or treat the non-IID split as a fixed setting rather than a variance component, and benchmarks published in 2025 and 2026 [8]–[10] still do so. Adjacent work decomposes variance over training or unlearning seeds and over benchmark variants [11]–[14], but not over the federated partition draw for LoRA instruction tuning. This paper’s advance is to isolate that draw in a pre-registered design, quantify its share relative to interaction and residual noise, and turn the measurement into an explicit reporting protocol that states how many draws were used, that comparisons were paired on those draws, and what minimum difference the design can detect. The same design also shows the cost of too few draws: extending LLaMA-3.2-3B from six to ten partitions reverses the FedIT–FLoRA top-two order and grows the interaction component by a factor of 2.66.

### B. CONTRIBUTIONS

(1) A pre-registered 204-run design at two scales, registered in two stages, that separates partition seeds from training seeds under a frozen federation and LoRA configuration. (2) A method-of-moments variance decomposition with cluster-bootstrap intervals, showing on TinyLlama that partition and interaction shares dominate the residual (0.508 and 0.469 versus 0.023), while on LLaMA-3.2-3B at  $p=10$  the interaction dominates (0.992) and the partition main-effect interval includes zero after truncation. (3) A ranking-stability analysis conditioned on effect size: the near-tied FedIT–FLoRA pair flips under few draws (three-draw paired

flip probability 0.377 on TinyLlama and 0.493 on LLaMA-3.2-3B), whereas gaps of about 0.023 to 0.031 never reverse in a single draw. (4) A draws-needed table for pre-registered deltas and for the observed gaps, including 206 and more than 1000 paired draws for the near-tie at the two scales, and the observation that where the interaction dominates, pairing on partitions buys little. (5) The Partition-Draw Reporting Protocol as three numbered steps (partition-draw count and seeds; paired comparisons; detectable minimum difference).

Section II situates the work against benchmark-variance studies, non-IID simulation practice, and federated LoRA evaluation. Section III states the design and variance model. Section IV reports the measurements. Section V gives reporting recommendations and limitations. Section VI concludes.

## II. RELATED WORK

### A. VARIANCE AND SEED SENSITIVITY IN ML BENCHMARKS

Empirical comparisons are sensitive to sources of variation beyond the method under test. Bouthillier et al. [11] model data sampling, initialization, and hyperparameter choice as variance components in machine-learning benchmarks and show that ignoring them inflates the chance of declaring a false improvement; their recommendations include randomizing multiple sources of variation and accounting for the resulting uncertainty when claiming gains. Picard [15] isolates the random seed alone: across 10,000 CIFAR-10 trainings the accuracy range ran from 89.01% to 90.83% (1.82 percentage points), with about 0.5% on ImageNet. Reimers and Gurevych [16] similarly argue for reporting score distributions rather than single scores when ranking methods. Agarwal et al. [12] show on Atari that few-run point estimates produce unreliable rankings, that the number of runs needed for tight intervals is far larger than common practice assumes, and that reporting intervals rather than point estimates changes which algorithm appears better. Their source of variation is the training run on a fixed task suite. The draw studied here is the data partition that defines the federation, which no amount of training-run replication resolves. These lines of work motivate multi-seed summaries and caution around small margins. Neither separates the draw of a federated data partition from the training seed, which is the decomposition this paper targets.

### B. NON-IID SIMULATION IN FEDERATED LEARNING

Federated evaluations routinely induce client heterogeneity by construction [17]. Hsu et al. [6] synthesize label skew by drawing each client’s class proportions from a Dirichlet distribution  $\text{Dir}(\alpha p)$ , with smaller  $\alpha$  producing more skewed clients, and show that FedAvg accuracy degrades as  $\alpha$  decreases. Natural partitions that follow writers, users, or categories appear in benchmarks such as LEAF [18]. NIID-Bench [7] compares algorithms under six partitioning strategies and reports that no existing algorithm outperforms the others in all cases. Federated instruction-tuning work inherits these simulation habits: FedIT [3] splits Databricks-dolly-

15k into ten shards, one per client, and reports per-client category distributions in which several categories are absent entirely, and FFA-LoRA [1] uses a 3-client cross-silo setting with fixed label proportions (for example [0.1, 0.9] and [0.9, 0.05, 0.05]), not a Dirichlet draw, on RoBERTa/GLUE and LLaMA/GSM-8K rather than Dolly. The practice is current rather than historical. ProFed [8] repeats each configuration over five seeds without stating whether a seed redraws the partition, and reports one spread per cell; a 2026 cross-domain benchmark for federated fine-tuning [9] fixes a single random seed and one Dirichlet partition per dataset and reports point estimates; and recent work on the reliability of federated evaluation [10] holds client data identical across models, so the partition is a constant rather than a draw. None of the evaluations above reports resampling the partition seed as a separate source of variation, and the single draw is treated as the non-IID condition, even though that draw determines which clients are empty of a category or too small to take an optimizer step.

### C. FEDERATED LORA METHODS AND EVALUATION PRACTICE

The federated LoRA literature is already large enough that evaluation practice matters for how methods are ranked [19]. FedIT [3] aggregates LoRA adapters with FedAvg for instruction tuning. FFA-LoRA [1] freezes  $A$  and trains  $B$  in settings drawn from FederatedScope-LLM [20], reporting accuracies over 20 runs as mean and standard deviation. In the non-private setting the MNLI-matched means are  $85.05 \pm 1.1$  (FFA-LoRA) versus  $82.03 \pm 10.7$  (LoRA), and the QQP means are  $84.35 \pm 0.6$  versus  $83.51 \pm 3.3$ ; the LoRA standard deviation of 10.7 on MNLI-matched exceeds the mean method gap, which is published evidence that run-to-run variance in federated LoRA can dominate the reported contrast. FLoRA [2] stacks heterogeneous-rank adapters, in contrast with zero-padding heterogeneous LoRA [21], and compares to FedIT on MMLU and MT-bench with 10 clients under FedIT’s non-IID setting (matching our client count). Several margins are modest: Llama/Alpaca MMLU is 29.85 versus 29.41, which the paper calls “marginal,” and homogeneous MT-bench gains are described as at least 0.2. Those runs use only 1 to 3 communication rounds with single reported values and no seed replication, and MMLU is scored on a 1,444-sample subset. Published evaluations therefore sometimes report across-run means and standard deviations, yet they do not resample the partition draw or decompose that run variance into partition and training-seed components when the reported margin is on the order of a point or less.

### D. GAP

Variance decomposition of evaluation noise is not new. Lanyon et al. [13] decompose unlearning-metric variance into training-seed and unlearning-seed terms and show that fixing a single training seed can misrepresent method variability. Chanin [14] reports ANOVA-style variance shares across SAE variants, training seeds, and training snapshots,

and shows that for several SAEbench metrics the crowned (best) variant flips across seeds. Rodriguez et al. [22] fix the data partition and vary only the training seed in recommender-system evaluation, reporting that seed variation is often detectable and that whether it changes model selection depends on how far apart the candidate configurations sit. In federated learning, the nearest work does resample the partition but pools the result into a single spread rather than decomposing it: Jimenez-Gutierrez et al. [23] run five and ten trials built from distinct data partitions with fixed random seeds and report mean accuracy with one standard deviation across trials, without separating the partition draw from the training seed or attributing variance to either; NIID-Bench [7] likewise varies an initial seed that controls both. What has not been done is to treat the federated partition draw itself as a variance component for LoRA instruction tuning, or to test whether method rankings survive that component. That is this paper’s case.

## III. EXPERIMENTAL SETUP

We study three federated LoRA aggregation methods that appear in recent instruction-tuning work: FedIT [3] (FedAvg [24] over LoRA adapters), FFA-LoRA [1], and FLoRA [2]. No other aggregators are included. The design is fixed before production runs and is identical across methods except for the aggregation rule itself. Table 1 summarizes the frozen design.

### A. MODELS AND ADAPTERS

The primary model is TinyLlama-1.1B-Chat-v1.0 [25]. The replication model is LLaMA-3.2-3B from the Llama 3 herd [26], retained after a pre-registered timing gate on an RTX 4090 measured 65.2 minutes for a full run with a peak allocation of 9.5 GB, within the gate’s limits of 75 minutes and 22 GB. Both models use the same LoRA [27] configuration: rank  $r=16$ , scaling  $\alpha_{\text{LoRA}}=32$ , dropout 0.1, and target modules restricted to  $q_{\text{proj}}$  and  $v_{\text{proj}}$ . The base model is frozen; only the LoRA parameters are trained and communicated. Both models use learning rate  $1 \times 10^{-4}$ , weight decay 0.01, and maximum sequence length 256.

### B. FEDERATION

Each run uses 10 clients with full participation for 15 rounds and one local epoch per round. TinyLlama uses batch size 4 with gradient accumulation 4; LLaMA-3.2-3B uses batch size 2 with gradient accumulation 8. The client training loop steps the optimizer only on complete gradient-accumulation blocks and does not use `drop_last`, so clients with fewer than 13 samples (TinyLlama) or 15 samples (LLaMA) contribute zero optimizer steps (an accumulation block is four micro-batches of size 4 on TinyLlama and eight of size 2 on LLaMA, and the final partial micro-batch counts toward a block). We call a client active if it holds at least one example, and effective if it also takes at least one optimizer step. In the Dirichlet  $\alpha=0.1$  partitions, effective clients range from 7 to 10 on TinyLlama and from 9 to 10 on LLaMA-3.2-3B;

TABLE 1. Frozen experimental design.

| Element            | Value                                                                               |
|--------------------|-------------------------------------------------------------------------------------|
| Methods            | FedIT, FFA-LoRA, FLoRA                                                              |
| Models             | TinyLlama-1.1B-Chat-v1.0; LLaMA-3.2-3B                                              |
| LoRA               | $r=16$ , $\alpha_{\text{LoRA}}=32$ , dropout 0.1, $q_{\text{proj}}+v_{\text{proj}}$ |
| Federation         | 10 clients, full participation, 15 rounds, 1 local epoch                            |
| Data               | Dolly-15k train[0:3000]; held-out train[3000:3500]                                  |
| Non-IID            | Dirichlet label skew on <code>category</code> , $\alpha=0.1$ and 0.5                |
| IID                | Fixed data seed split                                                               |
| Primary metric     | Held-out loss (formatter <code>v2-dolly-context</code> )                            |
| Secondary          | Measured communication (MB)                                                         |
| TinyLlama cells    | 60 ( $\alpha=0.1$ ) + 60 ( $\alpha=0.5$ ) + 15 (IID)                                |
| LLaMA-3.2-3B cells | 60 ( $\alpha=0.1$ , $p=10$ ) + 9 (IID)                                              |

discarded trailing samples per partition seed range from 27 to 60 (TinyLlama) and 41 to 104 (LLaMA); client sizes range from 2 to 1218; and active clients take only the values 9 or 10.

### C. DATA AND PARTITIONS

All training examples come from Databricks Dolly-15k [28], using `train[0:3000]`. The non-IID setting is Dirichlet label skew [6] over the dataset’s real `category` column at  $\alpha=0.1$ . The IID setting uses a single fixed data seed. Partition draws are controlled by data seeds that are separate from the training (run) seeds. TinyLlama uses data seeds 2001–2010 with run seeds {7001, 7002} at  $\alpha=0.1$  (60 cells), the same seed grid at  $\alpha=0.5$  (60 cells), and data seed 2001 with run seeds 7001–7005 for IID (15 cells). LLaMA-3.2-3B uses data seeds 2001–2010 with the same two run seeds at  $\alpha=0.1$  (60 cells) and data seed 2001 with run seeds 7001–7003 for IID (9 cells). The total design is 204 production cells. The first 120 cells (both models at  $\alpha=0.1$  with six LLaMA partitions, plus IID) were run under the original pre-registration; the remaining 84 extend TinyLlama to  $\alpha=0.5$  and LLaMA from six to ten partition draws, under an amended plan registered before those runs began. Existing federated LLM leaderboards and suites such as FlowerTune [29], OpenFedLLM [30], and FedLLM-Bench [31] report single-configuration results, which is the evaluation practice under study, so this design requires partition and training seeds under the authors’ control rather than a fixed public leaderboard split.

### D. HELD-OUT METRIC

The primary metric is held-out next-token loss on Dolly `train[3000:3500]` (500 examples), maximum sequence length 256, using the same prompt format as training, including a `Context` field when present (formatter version `v2-dolly-context`). Base (unadapted) held-out loss is 2.120666 on TinyLlama and 2.168946 on LLaMA-3.2-3B and is identical across all runs of a given model. TinyLlama holdout evaluation runs in float32. LLaMA-3.2-3B production holdouts run in float16 with sequential loading (the tuned adapter is freed before the base model is loaded) because

two concurrent float32 3B copies exceed 24 GB. A float32 spot-check on three LLaMA cells spanning the production tuned-loss range found a maximum absolute difference of  $4.1 \times 10^{-5}$  relative to float16, about 250 times below the 0.01 effect scale used for power calculations, so float16 was retained for the LLaMA production numbers.

### E. WHY HELD-OUT LOSS

Held-out next-token loss is the quantity the federated objective optimizes [24], and it is continuous, so it resolves differences that a thresholded accuracy on a finite evaluation set cannot. Partition-draw variance is a property of the measurement apparatus rather than of the metric: any quantity computed from a partitioned federation inherits it, so the decomposition reported here is not specific to the choice of response variable. Discretized task metrics on small evaluation sets carry their own sampling noise, which adds to partition variance rather than replacing it. For example, MMLU scored on 1,444 sampled questions at accuracy near 0.30 has a binomial standard error of about 1.2 percentage points per system. Because both systems are scored on the same questions, the standard error of their difference depends on their correlation; it is at most about 1.7 points under independent scoring and smaller when the systems agree on most items. Either way it is large relative to the 0.44-point margin reported for the Llama/Alpaca MMLU contrast in FLoRA [2]. Those standard-error figures are our calculation from the 1,444-sample size stated in that paper, not a quantity the paper reports.

### F. COMMUNICATION

Secondary reporting uses measured communication in megabytes, counting the tensors actually transmitted in each direction following the byte accounting in [32]. Per-client upload payloads match the frozen constants: 9,011,200 bytes for TinyLlama (FedIT and FLoRA) and 9,175,040 bytes for LLaMA-3.2-3B (FedIT and FLoRA). FFA-LoRA, as implemented in this codebase, uploads both *A* and *B* on the first round and *B* thereafter, and downloads *B* only (3,244,032 bytes per client on TinyLlama; 3,670,016 bytes per client on LLaMA-3.2-3B). Total communication therefore differs

across methods by construction. We report communication only within method across partitions and do not rank methods by bytes.

### G. HARDWARE AND SOFTWARE PROVENANCE

Every production cell ran on a single NVIDIA GeForce RTX 4090 with one worker process per GPU. Median wall-clock time is 27.1 minutes for TinyLlama and 51.1 minutes for LLaMA-3.2-3B. Training provenance is the `freeze-v1` tag for the original 120 cells and `freeze-v3.1` for the 84 extension cells, each with a clean tracked tree; the training path is byte-identical across those tags. The LLaMA float16 holdout change is an eval-only amendment tagged `freeze-v2`; it does not alter training code or aggregated adapters. The design, configurations, grids, and analysis plan were committed and tagged before any production run, and the analysis follows that plan; deviations are logged in the repository's decision log and reported here. Nine cells resumed from a checkpoint after host interruptions, and failed attempts were retried to completion. For the eight resumed cells with a sibling run seed, the held-out loss differs from that sibling by at most  $1.4 \times 10^{-3}$ , within the spread of sibling differences among cells that were not resumed (median  $3.1 \times 10^{-4}$ , maximum  $2.6 \times 10^{-3}$ ). The ninth, a LLaMA IID cell, differs from each of its other seeds by at most  $1.4 \times 10^{-3}$ .

### H. VARIANCE MODEL

For each model and heterogeneity level we fit a two-way design with methods fixed ( $m=3$ ), partitions random ( $p=10$ ;  $p=6$  is retained for LLaMA-3.2-3B as a pre-registered sensitivity analysis), and  $r=2$  training seeds per cell. Method-of-moments estimates [33] from the mean squares give the residual (training-seed) component  $\hat{\sigma}_E^2$ , the partition-by-method interaction  $\hat{\sigma}_{PM}^2$ , and the partition component  $\hat{\sigma}_P^2$ , with negative estimates truncated at zero and truncation reported. With  $p=10$  and  $m=3$  the residual component is pooled over 30 within-cell differences, and the IID cells supply three to five training seeds per method as an independent read on run-to-run noise. Confidence intervals come from a cluster bootstrap [34] that resamples partitions with replacement ( $B=2000$ ). A linear mixed-model fit (statsmodels MixedLM [35]) serves as a cross-check; the moment estimates are primary. Ranking stability uses  $B=10,000$  resamples of  $k$  partition draws under paired and unpaired protocols, and required sample sizes come from a two-sided  $t$  test at significance level 0.05 with 80% power. All random procedures use a fixed seed.

## IV. RESULTS

All analyses use held-out loss as the response. Numbers below come from the pre-registered tables in `analysis/`. We report TinyLlama and LLaMA-3.2-3B separately. Figure 1 shows the held-out loss of every cell by method and setting.

### A. VARIANCE COMPONENTS AT $\alpha=0.1$

Table 2 summarizes the method-of-moments estimates under the two-way design in Section III. Each share is a component divided by the total  $\hat{\sigma}_P^2 + \hat{\sigma}_{PM}^2 + \hat{\sigma}_E^2$ . On TinyLlama ( $p=10$ ), the partition share is 0.508 with bootstrap 95% CI [0.202, 0.663], the interaction share is 0.469 [0.316, 0.765], and the residual share is 0.023 [0.007, 0.056]. Absolute components are  $\hat{\sigma}_P^2 = 4.09 \times 10^{-6}$ ,  $\hat{\sigma}_{PM}^2 = 3.78 \times 10^{-6}$ , and  $\hat{\sigma}_E^2 = 1.89 \times 10^{-7}$ . Partition and interaction are comparable in magnitude; residual training-seed noise is an order of magnitude smaller. No estimate was truncated at zero. The pre-registered mixed-model cross-check recovers the residual component on both models ( $1.89 \times 10^{-7}$  and  $1.31 \times 10^{-7}$ , matching  $\hat{\sigma}_E^2$ ) and a method-level variance component comparable to  $\hat{\sigma}_P^2 + \hat{\sigma}_{PM}^2$ , but the separate partition component did not identify under the specified variance-component formula. We therefore report the moment estimates as primary, as pre-registered, and treat the mixed model as a partial cross-check.

On LLaMA-3.2-3B the primary analysis uses  $p=10$  partitions (Table 2). The interaction share is 0.992 [0.662, 0.995], with  $\hat{\sigma}_{PM}^2 = 2.94 \times 10^{-5}$  and residual share 0.008; the partition component truncates at zero (share 0.000, CI [0.000, 0.329]). Because that CI includes zero, we do not claim a partition main effect at 3B. Cross-scale bootstrap CIs on the partition share overlap ([0.202, 0.663] versus [0.000, 0.329]). The  $p=6$  sensitivity row is reported next.

### B. PARTITION COUNT AND RANKING (LLAMA-3.2-3B)

The LLaMA grid was extended from six partition draws to ten after the pre-registered design completed, which permits a direct test of how many draws an evaluation needs. At  $p=6$  the reference ranking placed FedIT ahead of FLoRA by 0.00201; at  $p=10$  the order reverses, with FLoRA ahead by 0.00067. Neither ordering is statistically distinguishable (Holm-adjusted [36]  $p=0.83$  at  $p=10$ ; the paired confidence interval  $[-0.00622, +0.00757]$  contains zero). The interaction component also grows from  $\hat{\sigma}_{PM}^2 = 1.11 \times 10^{-5}$  at  $p=6$  to  $2.94 \times 10^{-5}$  at  $p=10$ , a factor of 2.66, while the partition component truncates at zero (share 0.000, CI [0.000, 0.329], versus 0.227 [0.000, 0.551] at  $p=6$ ). An evaluation stopping at six draws would therefore have reported the opposite top-two ordering and understated the interaction variance by more than half. Six draws exceeds what the federated LoRA evaluations in Section II report. We treat  $p=10$  as primary and  $p=6$  as the pre-registered sensitivity analysis.

### C. A SECOND HETEROGENEITY LEVEL

Repeating the TinyLlama design at Dirichlet  $\alpha=0.5$  (10 partition draws, two training seeds, 60 cells) leaves the variance composition statistically unchanged: shares are 0.277 [0.000, 0.597], 0.682 [0.352, 0.958], and 0.042 [0.027, 0.094], and all three share differences against  $\alpha=0.1$  have bootstrap intervals containing zero ( $-0.231$  [ $-0.568, 0.194$ ] for the partition share). The pre-registered outcome is that the composition does not change detectably

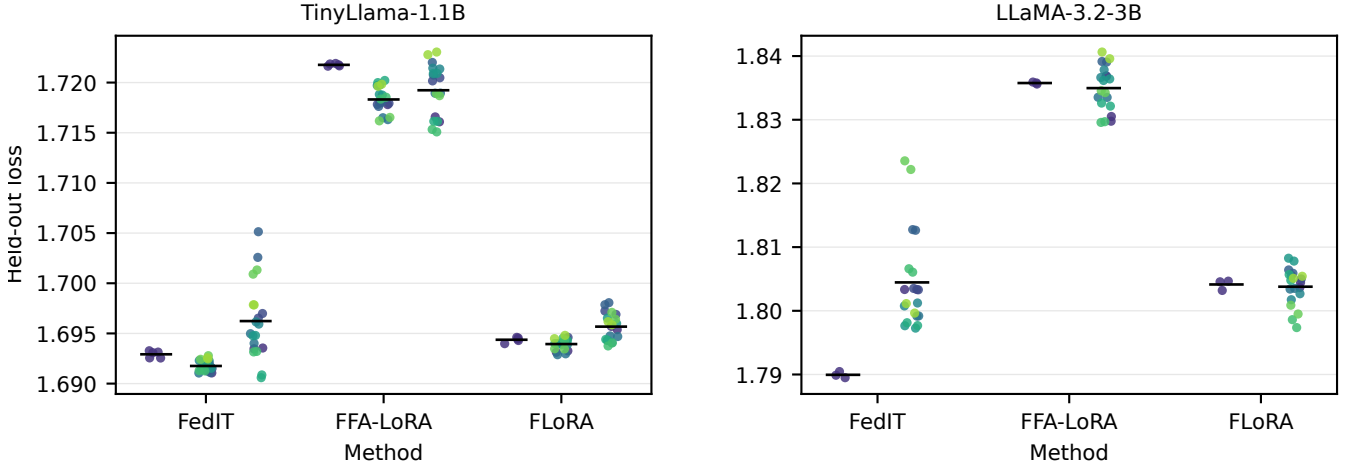


FIGURE 1. Held-out loss by method; clusters run left to right by setting (IID,  $\alpha=0.5$ ,  $\alpha=0.1$ ; LLaMA: IID,  $\alpha=0.1$ ). Bars mark means; colors mark partitions.

TABLE 2. Method-of-moments variance components by model and heterogeneity level, with bootstrap 95% CIs over partitions ( $B=2000$ ).

| Model          | Setting                 | $\hat{\sigma}_P^2$    | $\hat{\sigma}_{PM}^2$ | $\hat{\sigma}_E^2$    | Share $P$            | Share $PM$           | Share $E$            | $p$ | $m$ | $r$ |
|----------------|-------------------------|-----------------------|-----------------------|-----------------------|----------------------|----------------------|----------------------|-----|-----|-----|
| TinyLlama-1.1B | $\alpha=0.1$            | $4.09 \times 10^{-6}$ | $3.78 \times 10^{-6}$ | $1.89 \times 10^{-7}$ | 0.508 [0.202, 0.663] | 0.469 [0.316, 0.765] | 0.023 [0.007, 0.056] | 10  | 3   | 2   |
| TinyLlama-1.1B | $\alpha=0.5$            | $2.22 \times 10^{-7}$ | $5.47 \times 10^{-7}$ | $3.34 \times 10^{-8}$ | 0.277 [0.000, 0.597] | 0.682 [0.352, 0.958] | 0.042 [0.027, 0.094] | 10  | 3   | 2   |
| LLaMA-3.2-3B   | $\alpha=0.1, p=10$      | 0                     | $2.94 \times 10^{-5}$ | $2.51 \times 10^{-7}$ | 0.000 [0.000, 0.329] | 0.992 [0.662, 0.995] | 0.008 [0.005, 0.029] | 10  | 3   | 2   |
| LLaMA-3.2-3B   | $\alpha=0.1, p=6$ sens. | $3.28 \times 10^{-6}$ | $1.11 \times 10^{-5}$ | $1.31 \times 10^{-7}$ | 0.227 [0.000, 0.551] | 0.764 [0.443, 0.990] | 0.009 [0.002, 0.043] | 6   | 3   | 2   |

between these two skew levels. The absolute magnitudes do change:  $\hat{\sigma}_P^2$  falls from  $4.09 \times 10^{-6}$  to  $2.22 \times 10^{-7}$  and the paired per-draw SD from 0.00282 to 0.00108, so total variance is about an order of magnitude smaller at  $\alpha=0.5$ . This magnitude comparison was not pre-registered and is reported as an observation. One consequence is visible in the contrasts: the FedIT–FLoRA pair, indistinguishable at  $\alpha=0.1$ , separates at  $\alpha=0.5$  ( $-0.00218$ , Holm  $p=3.2 \times 10^{-6}$ ). All ten clients are active in every  $\alpha=0.5$  partition, so measured communication is constant within method.

#### D. IID NOISE FLOOR

Pooling the per-method IID variances gives  $6.40 \times 10^{-8}$  on TinyLlama and  $3.07 \times 10^{-7}$  on LLaMA-3.2-3B. The ratio of  $\hat{\sigma}_E^2$  at  $\alpha=0.1$  to this IID floor is 2.96 on TinyLlama. On LLaMA the ratio is 0.43 computed against the six-draw residual, which is the fit the pre-registered noise-floor comparison is defined against; the ten-draw residual gives 0.82. Each method’s IID variance has only two degrees of freedom (three run seeds), and the per-method IID variances differ by more than an order of magnitude ( $2.4 \times 10^{-8}$  for FFA-LoRA versus  $6.6 \times 10^{-7}$  for FLoRA). We therefore report the 3B IID floor descriptively and do not read either ratio below one as evidence that  $\alpha=0.1$  is quieter than IID.

#### E. RANKING STABILITY

The reference ranking at  $\alpha=0.1$  orders methods by mean held-out loss over all cells: TinyLlama ranks FLoRA, FedIT, then FFA-LoRA; LLaMA-3.2-3B at  $p=10$  ranks FLoRA,

FedIT, then FFA-LoRA. Under the pre-registered paired protocol ( $B=10,000$ ), the probability that the best method differs from this reference equals the probability that the full order differs (the two coincide because the only pair whose order changes is the top one; the third method is separated from both by about four (LLaMA-3.2-3B) to eight (TinyLlama) times the per-draw paired SD). On TinyLlama those probabilities are 0.555, 0.406, 0.377, and 0.284 for  $k \in \{1, 2, 3, 5\}$  (Wilson [37] 95% CI half-widths about 0.01). On LLaMA-3.2-3B at  $p=10$  they are 0.700, 0.533, and 0.493 for  $k \in \{1, 2, 3\}$  (unpaired 0.599, 0.528, 0.483), so a three-draw evaluation at 3B is close to a coin flip on the top two. Unpaired probabilities are shown in Figure 2; the paired protocol is primary.

The aggregate flip rate is entirely the near-tied FedIT–FLoRA pair. On TinyLlama that pair has full-design mean gap  $+0.00055$ ; single-draw sign flips occur in 11 of 20 cells, and the paired order-flip probability is 0.555 at  $k=1$  and 0.377 at  $k=3$ . The separable pairs (FedIT versus FFA-LoRA at  $-0.023$ , FFA-LoRA versus FLoRA at  $+0.024$ ) never reverse: 0 of 20 single draws and paired flip probability 0 at every  $k$ . On LLaMA-3.2-3B, FedIT versus FLoRA has gap  $+0.00067$  with a  $k=1$  paired flip probability of 0.700, while the separable gaps ( $-0.031$  and  $+0.031$ ) never reverse at any  $k$ .

Conditioned on effect size, the ranking claim is therefore: pairs separated by less than the per-draw paired SD (0.00282 on TinyLlama, 0.00770 on LLaMA-3.2-3B at  $p=10$ ) are ordered near-randomly by a few-partition evaluation, while

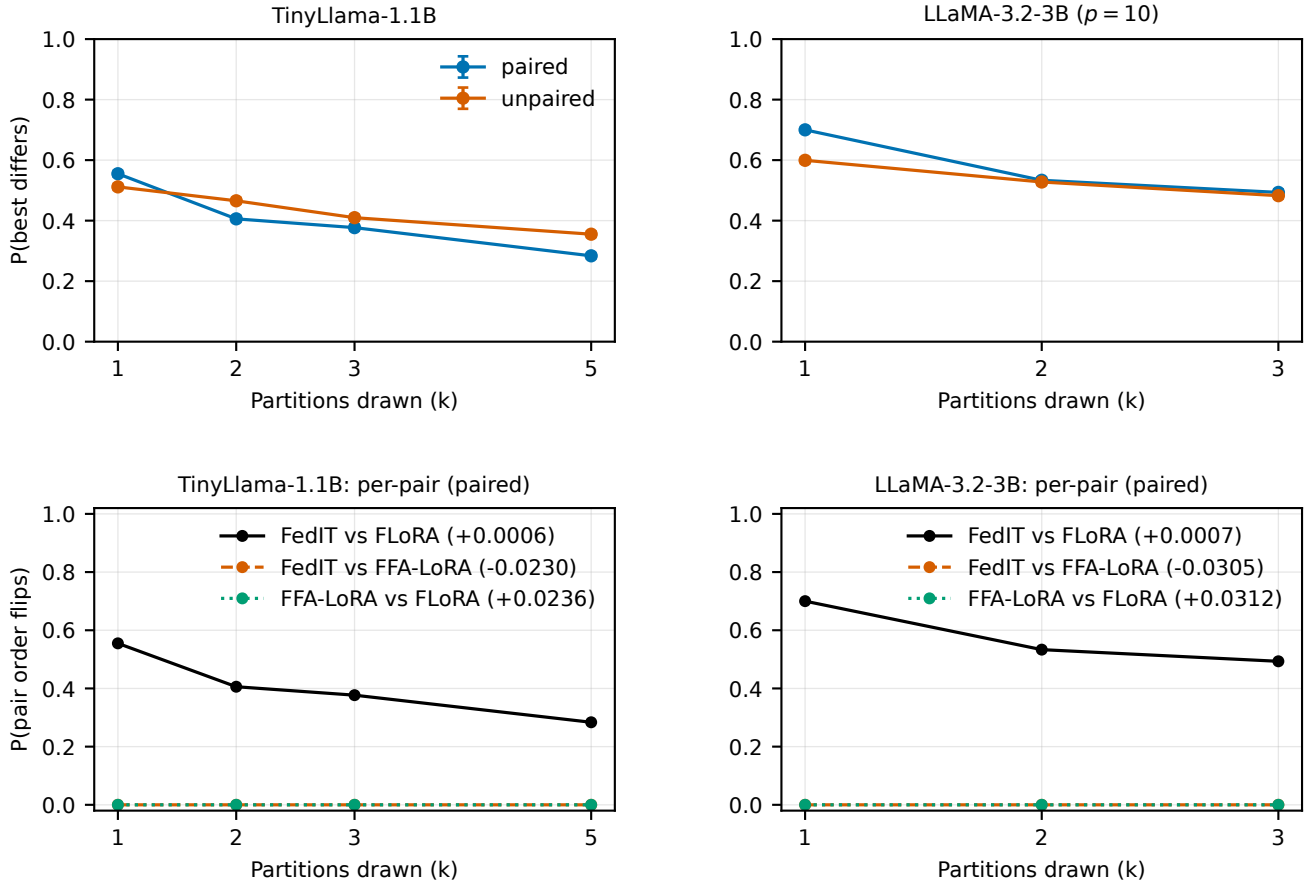


FIGURE 2. Ranking-flip probabilities under paired and unpaired protocols.

gaps of about 0.023 (TinyLlama) and 0.031 (LLaMA) were recovered in every single draw. The flip table is a restatement of that conditioning, not a separate claim about method quality.

#### F. DRAWS NEEDED

Using the model-based paired SD (Figure 3), detecting a pre-registered  $\Delta=0.005$  at 80% power (two-sided, significance level 0.05) requires 5 paired partition draws on TinyLlama. At 3B with ten draws,  $\Delta=0.005$  requires 21 paired draws,  $\Delta=0.01$  requires 7,  $\Delta=0.02$  requires 4, and  $\Delta=0.05$  requires 3 (unpaired per method: 20, 6, 3, 2). For the observed gaps, each separable pair needs 3 paired draws, whereas the FedIT-FLoRA near-tie needs more than 1000 under either protocol. The 1.1B near-tie needs 206 paired draws. That contrast is the operational form of the ranking result: a few partitions suffice when the gap is large relative to the paired SD, and hundreds to more than a thousand are required when it is not.

#### G. OBSERVED METHOD DIFFERENCES

Table 3 reports means and SDs. At  $\alpha=0.1$ , TinyLlama means are FLoRA 1.6957 (SD 0.0013), FedIT 1.6962 (0.0038), and

FFA-LoRA 1.7192 (0.0026), each over  $n=20$  cells; LLaMA-3.2-3B at  $p=10$  means are FLoRA 1.8038 (0.0029), FedIT 1.8045 (0.0077), and FFA-LoRA 1.8350 (0.0035), each over  $n=20$ . Paired  $t$ -tests on partition-level means with Holm correction [36] reject FFA-LoRA versus each of the other two methods on both models (TinyLlama Holm  $p<10^{-8}$ ; LLaMA Holm  $p<10^{-5}$ ) and do not reject FedIT versus FLoRA at  $\alpha=0.1$  (TinyLlama Holm  $p=0.57$ ; LLaMA Holm  $p=0.83$ ). The FedIT-FLoRA confidence intervals include zero on both models at  $\alpha=0.1$ . We report these contrasts as context for the ranking and power sections and do not claim a winning method.

#### H. PARTITION STATISTICS AND COMMUNICATION

Among five exploratory Spearman correlations of partition-mean held-out loss with partition statistics, TinyLlama mean Jensen-Shannon divergence [38] has  $\rho=0.72$  ( $p=0.019$ ,  $n=10$ ). That association does not survive Holm correction across the five TinyLlama tests (Holm  $p=0.093$ ) and does not replicate on LLaMA ( $\rho=0.086$ ,  $p=0.87$ ,  $n=6$ ). Active clients vary only from 9 to 10 at  $\alpha=0.1$  and show no reliable association ( $\rho=0.35$ ,  $p=0.32$  on TinyLlama;  $\rho=0.13$ ,  $p=0.80$  on LLaMA). We treat RQ4 as exploratory and do not build an

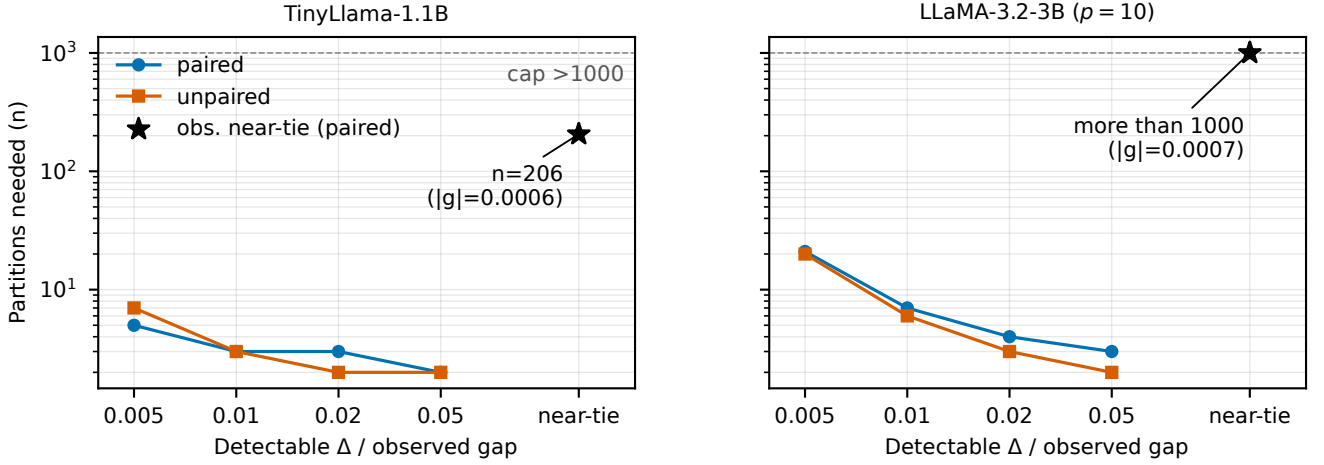


FIGURE 3. Partition draws needed for pre-registered deltas and the observed FedIT-FLoRA near-tie (star).

TABLE 3. Method means and communication ranges by model and setting.

| Model                           | Method   | Mean   | SD     | Comm. range (MB) | Active clients |
|---------------------------------|----------|--------|--------|------------------|----------------|
| TinyLlama-1.1B ( $\alpha=0.1$ ) | FedIT    | 1.6962 | 0.0038 | 2320.3–2578.1    | 9–10           |
| TinyLlama-1.1B ( $\alpha=0.1$ ) | FFA-LoRA | 1.7192 | 0.0026 | 884.8–983.1      | 9–10           |
| TinyLlama-1.1B ( $\alpha=0.1$ ) | FLoRA    | 1.6957 | 0.0013 | 2320.3–2578.1    | 9–10           |
| TinyLlama-1.1B ( $\alpha=0.5$ ) | FedIT    | 1.6918 | 0.0006 | 2578.1           | 10             |
| TinyLlama-1.1B ( $\alpha=0.5$ ) | FFA-LoRA | 1.7183 | 0.0013 | 983.1            | 10             |
| TinyLlama-1.1B ( $\alpha=0.5$ ) | FLoRA    | 1.6939 | 0.0006 | 2578.1           | 10             |
| LLaMA-3.2-3B ( $p=10$ )         | FedIT    | 1.8045 | 0.0077 | 2362.5–2625.0    | 9–10           |
| LLaMA-3.2-3B ( $p=10$ )         | FFA-LoRA | 1.8350 | 0.0035 | 992.3–1102.5     | 9–10           |
| LLaMA-3.2-3B ( $p=10$ )         | FLoRA    | 1.8038 | 0.0029 | 2362.5–2625.0    | 9–10           |

argument on any single correlation.

Measured total communication matches the per-client payload constants of Section III, multiplied by active clients and rounds, exactly for all method–model pairs ( $\max |\text{error}|=0$ ). Across  $\alpha=0.1$  partitions, TinyLlama totals range from 2320.3 to 2578.1 MB for FedIT and FLoRA and from 884.8 to 983.1 MB for FFA-LoRA; LLaMA ranges are 2362.5 to 2625.0 MB and 992.3 to 1102.5 MB. At  $\alpha=0.5$ , every TinyLlama partition has ten active clients, so communication is constant within method (2578.1 MB for FedIT/FLoRA; 983.1 MB for FFA-LoRA). The FFA-LoRA range is lower because this implementation uploads  $A$  and  $B$  in the first round and  $B$  alone thereafter, and downloads  $B$  only (Section III); we report the values as measured within method and do not rank methods by bytes.

### I. PRE-REGISTERED CLAIM MAPPING

Following the outcome-to-claim tables in the repository analysis plan (`docs/ANALYSIS_PLAN.md`), TinyLlama meets the I7 condition “Partition share at least 30%, or 3-draw paired flip probability at least 20%” on the partition share (0.508). At  $p=10$ , LLaMA meets the I8–I10 condition “At  $p=10$ , the LLaMA partition-share CI still includes zero,”

with a truncated partition component whose CI includes zero; the three-draw flip probability is 0.493. The  $\alpha=0.5$  share comparison selects the I8–I10 condition “Gradient absent: all three difference CIs include zero.” Cross-scale partition-share CIs overlap, so the data cannot distinguish the two scales on the partition share.

## V. DISCUSSION

### A. THE PARTITION-DRAW REPORTING PROTOCOL

A multi-seed mean alone does not answer whether a federated LoRA ranking is stable under the non-IID draw. FFA-LoRA already shows that across-run standard deviations can exceed method gaps [1]; our results go one step further by decomposing that variation into a partition component, a partition-by-method interaction, and a residual training-seed component. At Dirichlet  $\alpha=0.1$ , the residual  $\hat{\sigma}_E^2$  is small relative to  $\hat{\sigma}_P^2$  and  $\hat{\sigma}_{PM}^2$  on TinyLlama, so reporting only run seeds understates the uncertainty that matters for ranking. We therefore propose the Partition-Draw Reporting Protocol (hereafter, the protocol):

- 1) Report the number of partition draws, and the partition seeds, separately from training seeds.
- 2) Pair method comparisons on the same partition draws.

- 3) Report the minimum difference the design can detect, at the design's number of draws.

When the observed gap is near the paired per-draw SD (0.00282 on TinyLlama, 0.00770 on LLaMA-3.2-3B at  $p=10$ ), a few-draw evaluation is nearly a coin flip; when the gap is about 0.023 to 0.031, three paired draws sufficed here. The six-to-ten-draw reversal at 3B is the same point demonstrated inside the experiment: stopping at six draws would reverse the top two and cut the estimated interaction variance by more than half.

### B. WHAT PAIRING FIXES, AND WHAT IT DOES NOT

Pairing methods on the same partition draws removes the shared partition main effect from the comparison and is the right default for ranking. It does not remove the interaction: methods can still reorder across draws when their responses to a partition differ. That is exactly the FedIT-FLoRA near-tie (+0.00055 on TinyLlama, +0.00067 on LLaMA-3.2-3B at  $p=10$ ), where best-method and full-order flip probabilities coincide because only the top pair moves. Pairing also does not invent power. Detecting a pre-registered  $\Delta=0.005$  still needs 5 (TinyLlama) or 21 (LLaMA-3.2-3B at  $p=10$ ) paired draws, and the observed near-tie would need 206 and more than 1000. For that near-tie, pairing roughly halves the requirement at 1.1B (206 versus 416 per method unpaired) but at 3B both protocols exceed 1000. At some larger  $\Delta$  the unpaired design needs fewer draws per method, because the interaction term enters the paired difference twice; where  $\hat{\sigma}_{PM}^2$  dominates, pairing on partitions buys little. Pairing is a variance-control device for the shared partition effect, not a substitute for enough independent partitions.

### C. CROSS-SCALE READING

On TinyLlama the partition share is substantial (0.508). On LLaMA-3.2-3B at  $p=10$  the interaction dominates (0.992) and the partition main-effect CI includes zero after truncation, so we report no partition main effect at 3B. The draw still governs the result at 3B: it acts through the partition-by-method interaction rather than through a component shared by all three methods, which is the harder case for a few-draw evaluation because pairing cannot remove it. The cross-scale CIs on the partition share overlap, and the two scales differ in more than parameter count: TinyLlama-1.1B-Chat-v1.0 is instruction-tuned [25] while LLaMA-3.2-3B is a base model [26], and the base weights are float32 against float16. The design was therefore scoped to measure the partition component within each scale rather than to attribute differences between them, and the 3B IID floor is reported descriptively at 2 degrees of freedom per method. The within-scale conclusions do not depend on a scale comparison. We make no method recommendation: the separable FFA-LoRA gaps are large and stable, the FedIT-FLoRA gap is not, and the contribution is the measurement design and the conditioning of ranking stability on effect size relative to partition-draw noise.

### D. SCOPE AND LIMITATIONS

The design covers two model scales (1.1B and 3B), one dataset (Dolly), two heterogeneity levels at 1.1B ( $\alpha=0.1$  and 0.5), three aggregators, 10 clients with full participation, 15 rounds with one local epoch, and held-out next-token loss as the single response. Generation benchmarks, additional datasets, partial participation, and larger scales lie outside it, so the reported shares describe this apparatus rather than federated LoRA in general. One property of the frozen federation, stated in Section III, shapes what the partition draw varies: local updates apply only on complete gradient-accumulation blocks, so the smallest clients at  $\alpha=0.1$  contribute no optimizer step. Communication is reported within method only, and RQ4 is exploratory and is not used for claims. The mechanism that produces the partition component is not specific to these settings. A Dirichlet draw over  $C$  categories and  $K$  clients fixes which clients hold too few examples to take an optimizer step and which categories are absent from a client, and that assignment is a combinatorial property of  $(\alpha, C, K)$  rather than of model size, corpus size, or round count. A design with small  $\alpha$  relative to  $C/K$  should therefore carry a partition component whatever the scale, which is what makes the reporting protocol worth applying beyond the conditions measured here.

Two measurement choices carry quantified bounds. LLaMA production holdouts run in float16; the float32 spot-check in Section III bounds the difference at  $4.1 \times 10^{-5}$ , about 250 times below the 0.01 effect scale used for power, so the 3B numbers are reported in float16. The mixed-model cross-check recovers the residual and method-level components but not a separate partition component, as reported in Section IV; the method-of-moments estimates are primary, as pre-registered.

The campaign's operational record is complete in the repository's decision log, and the two items that touch the reported numbers are summarized here because measurement discipline is the paper's subject. The held-out evaluation was amended mid-campaign to restore float32 for TinyLlama (freeze-v3.1), and 15 LLaMA held-out evaluations were recomputed so that every reported value comes from one code path; the TinyLlama base loss is identical to  $10^{-6}$  across all freeze tags, so the amendment leaves the reported values unchanged. Separately, 54 extension cells trained on hosts that installed a torch build other than the pinned 2.2.0+cu121 and were retrained on the pinned stack, with the originals retained for comparison. That comparison is itself a measurement: the stack shift averages  $-8.5 \times 10^{-8}$  at 1.1B and  $+8.4 \times 10^{-4}$  at 3B (SD  $3.0 \times 10^{-4}$ ,  $n=15$ ), so at 3B an undeclared component of the apparatus moved the metric by more than the FedIT-FLoRA gap of 0.00067 under test. This is the paper's argument arriving from the inside: components that go unreported are not thereby small. All reported results use the pinned stack, and the release covers results and metadata but not adapter weights.

## VI. CONCLUSION

At Dirichlet  $\alpha=0.1$ , the partition draw and its interaction with the method are first-order sources of uncertainty in federated LoRA evaluations. On TinyLlama the partition draw alone accounts for about half of held-out-loss variance, with a comparable interaction share; on LLaMA-3.2-3B at ten partition draws the interaction dominates and the partition main-effect interval includes zero after truncation. Extending that 3B grid from six to ten draws reverses the top-two FedIT-FLoRA ordering and multiplies the interaction component by 2.66, so an evaluation that stopped at six draws would report the opposite ranking and understate interaction variance. Rankings of near-tied methods flip under few draws, while gaps four to eight times the per-draw paired SD do not. Detecting the observed FedIT-FLoRA near-tie would require hundreds of paired partitions at 1.1B and more than a thousand at 3B. Evaluations should therefore report how many partition draws were used, pair methods on the same draws, and state the smallest difference the design can detect. We make no method recommendation.

## DECLARATIONS

### DATA AND CODE AVAILABILITY

Code, configurations, the pre-registered analysis plan, per-run metadata, and analysis outputs are available at [github.com/jerryadamsfranklin/fedlora-partition-variance](https://github.com/jerryadamsfranklin/fedlora-partition-variance) and archived at <https://doi.org/10.5281/zenodo.22861074>, which resolves to the most recent archived version. The submitted snapshot is release v1.1.0. Adapter weights are not included in the archive.

### FUNDING

None.

### CONFLICTS OF INTEREST

None.

### GENERATIVE AI

The author used Claude (Anthropic) for language editing and drafting support, and AI coding assistants for implementing experiment and analysis scripts. The author designed the study, verified all code and results, and is responsible for all technical content, analysis, and interpretations.

## REFERENCES

- [1] Y. Sun, Z. Li, Y. Li, and B. Ding, "Improving LoRA in privacy-preserving federated learning," in The Twelfth International Conference on Learning Representations, 2024. [Online]. Available: <https://openreview.net/forum?id=NLPzL6HWNl>
- [2] Z. Wang, Z. Shen, Y. He, G. Sun, H. Wang, L. Lyu, and A. Li, "FLoRA: Federated fine-tuning large language models with heterogeneous low-rank adaptations," in Advances in Neural Information Processing Systems, vol. 37, 2024, pp. 22 513–22 533.
- [3] J. Zhang, S. Vahidian, M. Kuo, C. Li, R. Zhang, T. Yu, G. Wang, and Y. Chen, "Towards building the FederatedGPT: Federated instruction tuning," in ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2024, pp. 6915–6919.
- [4] D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt, "Measuring massive multitask language understanding," in International Conference on Learning Representations, 2021. [Online]. Available: <https://openreview.net/forum?id=d7KBjml3GmQ>
- [5] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica, "Judging LLM-as-a-judge with MT-Bench and Chatbot Arena," in Advances in Neural Information Processing Systems 36: Datasets and Benchmarks Track, 2023.
- [6] T.-M. H. Hsu, H. Qi, and M. Brown, "Measuring the effects of non-identical data distribution for federated visual classification," arXiv preprint arXiv:1909.06335, 2019.
- [7] Q. Li, Y. Diao, Q. Chen, and B. He, "Federated learning on non-IID data silos: An experimental study," in 2022 IEEE 38th International Conference on Data Engineering (ICDE), 2022, pp. 965–978.
- [8] D. Domini, G. Aguzzi, and M. Viroli, "ProFed: A benchmark for proximity-based non-IID federated learning," arXiv preprint arXiv:2503.20618, 2025.
- [9] D. M. Jimenez-Gutierrez, E. Zuazua, G. Kellaris, J. del Rio, O. Slusarenko, and X. Uribe-Etxebarria, "Towards the next frontier of LLMs, training on private data: A cross-domain benchmark for federated fine-tuning," arXiv preprint arXiv:2605.13936, 2026.
- [10] C. Grosser, M. Heuer, D. Krompass, and T. A. Runkler, "Evaluating federated pre-training: On the reliability of downstream fine-tuning and intrinsic evaluation," arXiv preprint arXiv:2607.28658, 2026.
- [11] X. Bouthillier, P. Delaunay, M. Bronzi, A. Trofimov, B. Nichyporuk, J. Szeto, N. Mohammadi Sepahvand, E. Raff, K. Madan, V. Voleti, S. Ebrahimi Kahou, V. Michalski, T. Arbel, C. Pal, G. Varoquaux, and P. Vincent, "Accounting for variance in machine learning benchmarks," Proceedings of Machine Learning and Systems, vol. 3, pp. 747–769, 2021.
- [12] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, and M. G. Bellemare, "Deep reinforcement learning at the edge of the statistical precipice," in Advances in Neural Information Processing Systems, vol. 34, 2021.
- [13] J. Lanyon, A. Finke, P. Andreou, and G. Cosma, "On the importance of multiple training seeds for evaluating machine unlearning," arXiv preprint arXiv:2510.26714, 2025.
- [14] D. Chanan, "Are sparse autoencoder benchmarks reliable?" arXiv preprint arXiv:2605.18229, 2026.
- [15] D. Picard, "torch.manual\_seed(3407) is all you need: On the influence of random seeds in deep learning architectures for computer vision," arXiv preprint arXiv:2109.08203, 2021.
- [16] N. Reimers and I. Gurevych, "Reporting score distributions makes a difference: Performance study of LSTM-networks for sequence tagging," in Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing, 2017, pp. 338–348.
- [17] P. Kairouz, H. B. McMahan et al., "Advances and open problems in federated learning," Foundations and Trends in Machine Learning, vol. 14, no. 1–2, pp. 1–210, 2021.
- [18] S. Caldas, S. M. K. Duddu, P. Wu, T. Li, J. Konečný, H. B. McMahan, V. Smith, and A. Talwalkar, "LEAF: A benchmark for federated settings," arXiv preprint arXiv:1812.01097, 2018.
- [19] Y. Yang, G. Long, Q. Lu, L. Zhu, J. Jiang, and C. Zhang, "Federated low-rank adaptation for foundation models: A survey," in Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI-25), 2025, pp. 10 779–10 787.
- [20] W. Kuang, B. Qian, Z. Li, D. Chen, D. Gao, X. Pan, Y. Xie, Y. Li, B. Ding, and J. Zhou, "FederatedScope-LLM: A comprehensive package for fine-tuning large language models in federated learning," arXiv preprint arXiv:2309.00363, 2023.
- [21] Y. J. Cho, L. Liu, Z. Xu, A. Fahrezi, and G. Joshi, "Heterogeneous LoRA for federated fine-tuning of on-device foundation models," in Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, 2024, pp. 12 903–12 913.
- [22] J. M. Rodriguez, O. Lesota, and A. Tommasel, "Training seeds and model-selection stability in recommender-system evaluation," arXiv preprint arXiv:2609.02499, 2026, accepted at ACM RecSys 2026, Research and Practice Notes.
- [23] D. M. Jimenez-Gutierrez, M. Hassanzadeh, A. Anagnostopoulos, I. Chatzigiannakis, and A. Vitaletti, "A thorough assessment of the non-IID data impact in federated learning," Journal of Industrial Information Integration, vol. 50, p. 101052, 2026.
- [24] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Aguera y Arcas, "Communication-efficient learning of deep networks from decentralized data," in Proceedings of the 20th International Conference on Artificial In-

- telligence and Statistics, ser. Proceedings of Machine Learning Research, vol. 54. PMLR, 2017, pp. 1273–1282.
- [25] P. Zhang, G. Zeng, T. Wang, and W. Lu, “TinyLlama: An open-source small language model,” arXiv preprint arXiv:2401.02385, 2024.
  - [26] A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur et al., “The Llama 3 herd of models,” arXiv preprint arXiv:2407.21783, 2024.
  - [27] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-rank adaptation of large language models,” in International Conference on Learning Representations, 2022. [Online]. Available: <https://openreview.net/forum?id=nZeVKeeFYf9>
  - [28] M. Conover, M. Hayes, A. Mathur, J. Xie, J. Wan, S. Shah, A. Ghodsi, P. Wendell, M. Zaharia, and R. Xin, “Free Dolly: Introducing the world’s first truly open instruction-tuned LLM,” Databricks blog, 2023. [Online]. Available: <https://www.databricks.com/blog/2023/04/12/dolly-first-open-commercially-viable-instruction-tuned-llm>
  - [29] Y. Gao, M. R. Scamarcia, J. Fernandez-Marques, M. Naseri, C. S. Ng, D. Stripelis, Z. Li, T. Shen, J. Bai, D. Chen, Z. Zhang, R. Hu, I. Song, K. Lee, H. Jia, T. Dang, J. Wang, Z. Liu, D. J. Beutel, L. Lyu, and N. D. Lane, “FlowerTune: A cross-domain benchmark for federated fine-tuning of large language models,” arXiv preprint arXiv:2506.02961, 2025.
  - [30] R. Ye, W. Wang, J. Chai, D. Li, Z. Li, Y. Xu, Y. Du, Y. Wang, and S. Chen, “OpenFedLLM: Training large language models on decentralized private data via federated learning,” in Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2024, pp. 6137–6147.
  - [31] R. Ye, R. Ge, X. Zhu, J. Chai, Y. Du, Y. Liu, Y. Wang, and S. Chen, “FedLLM-Bench: Realistic benchmarks for federated learning of large language models,” in Advances in Neural Information Processing Systems 37: Datasets and Benchmarks Track, 2024, pp. 111 106–111 130.
  - [32] J. A. Franklin, “Adaptive phase-switching for communication-efficient federated LoRA fine-tuning,” arXiv preprint arXiv:2609.13512, 2026.
  - [33] S. R. Searle, G. Casella, and C. E. McCulloch, Variance Components. Wiley, 1992.
  - [34] C. A. Field and A. H. Welsh, “Bootstrapping clustered data,” Journal of the Royal Statistical Society: Series B (Statistical Methodology), vol. 69, no. 3, pp. 369–390, 2007.
  - [35] S. Seabold and J. Perktold, “Statsmodels: Econometric and statistical modeling with Python,” in Proceedings of the 9th Python in Science Conference, 2010, pp. 92–96.
  - [36] S. Holm, “A simple sequentially rejective multiple test procedure,” Scandinavian Journal of Statistics, vol. 6, no. 2, pp. 65–70, 1979.
  - [37] E. B. Wilson, “Probable inference, the law of succession, and statistical inference,” Journal of the American Statistical Association, vol. 22, no. 158, pp. 209–212, 1927.
  - [38] J. Lin, “Divergence measures based on the Shannon entropy,” IEEE Transactions on Information Theory, vol. 37, no. 1, pp. 145–151, 1991.



JERRY ADAMS FRANKLIN received the M.S. degree in data science from Northeastern University, Boston, MA, USA. He has worked as a deep learning engineer at Intel and as a senior AI/ML engineer at Digital Currency Group. His research interests include federated learning and resource-efficient fine-tuning of large language models.

...